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SEPM Assignment-2

Q1 Explain system development life cycle in detail

The system development life cycle (SDLC) is a structured process used by IT entities to design, develop and test-high-quality software. It provides a systematic sequence of stages that help an organization deliver a product that is cost-effective and meets user requirements.

Stages of SDLC

- Planning and requirement analysis: The most crucial stage where senior members gather inputs from customers, sales and market surveys. This defines the scope and identifies risks.
- Defining requirements: All requirements are documented in an SRS (Software requirement specification) document, which serves as a guide for the next phases.
- Design: Architects create the software architecture based on the SRS. This includes internal design (database, modules) and external design (UI/UX).
- Development (coding): This is the longest phase where the actual code is written by developers.

according to the design specifications.

→ **Testing** - The software is tested for bugs and errors. Testers verify if the application works according to the original requirements.

→ **Deployment & maintenance**: Once the software is bug-free, it is released to the market. Maintenance involves fixing any issues discovered by users and updating the software to meet changing needs.

Q2 Define a project? What is project management?
Explain various characteristics of project management.

Definition of a project: A project is a temporary endeavor undertaken to create a unique product, service or result. Unlike "operations," which are ongoing and repetitive, a project has a specific start date and end date.

Project Management (PM)

Project Management is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements. It aims to balance the "Triple Constraint": scope, time, and cost.

Characteristics of Project Management

- Temporary: It has a defined beginning and end
- Unique: Every Project produces a deliverable that is distinct from previous ones
- Progressive Elaboration: Projects are often defined broadly at the start and become more detailed as the team gains a better understanding of the goals.
- Single point of responsibility: Usually led by a project manager who is accountable for success.
- Multi-disciplinary: often involves people from different departments or specialities working together.

Q3 Write a short note on PMBOK

- The Project Management Body of Knowledge (PMBOK) is a set of standard terminology and guidelines for project management published by the Project Management Institute (PMI)
- Global Standard: It is recognized worldwide as the "bible" of project management.

- The 5 process groups: It organises project activities into initiating, planning, executing, monitoring & controlling and closing.
- Knowledge Areas: It covers 10 key areas including integration, scope, schedule, cost, quality, resources, communications, risk, procurement and stakeholder management.
- Standard Vocabulary: PMBOK provides a common language for professionals globally, ensuring that a "work breakdown structure" or "critical path" means the same thing to the managers to companies belonging to different cities in the different countries.
- Q4 → Evaluation: It is periodically updated to move from a process-based approach to a principle-based approach, reflecting modern agile and hybrid work environments.

Q4 Explain ITPM?

- ITPM is the specialized application of project management principles to the field of information Technology. While general project management can apply to construction or manufacturing, ITPM deals with software development, hardware installations, network upgrades, and data management.

why ITPM is unique.

- Intangibility: Unlike a bridge or a building, software is "invisible." This makes it harder to measure progress and easier for requirements to change.
- High complexity and Integration: IT projects often must integrate with existing legacy systems, various databases, and different hardware platforms.
- Rapid obsolescence: Technology changes so fast that the tools or versions used at the start of the project might be outdated by the time it finishes.

The ITPM Lifecycle:

- ITPM combines the Project management lifecycle (Initiating, Planning, Executing, Closing) with the system development cycle (SDLC).

Key Focus areas in ITPM.

- Scope management: Preventing "scope creep", where users keep adding new features to software, causing delays and budget overruns.
- Risk Management: IT projects face unique risks like cybersecurity threats, data loss, and software bugs.

- Quality Management: Ensuring the final product is user-friendly, scalable, and meets performance benchmarks.
- Stakeholder communication: Bridging the gap between "technical" developers and "non-technical" business clients to ensure the final product actually solves a business problem.

Q5 What is the role of RFP? Describe its content and purpose?

- A Request for proposal (RFP) is a document that an organization puts to elicit bids from potential vendors for a product or service.

Purpose:

- To notify vendors that the organization is looking to procure something
- To provide a level playing field for vendors to compete.
- To gather technical expertise and pricing from outside experts.

Contents of an RFP:

- Executive summary: overview of the project.
- Scope of work: Detailed description of what the vendor is expected to deliver.
- Technical Requirements: Specific hardware or ~~some~~ software standards.
- Timeline: Expected milestones and completion date.
- Evaluation Criteria: How the winner will be chosen.
- Submission guidelines: Format and deadline for the proposal.

Q 6 What is feasibility study? Explain its types, contents and purpose?

→ A Feasibility Study is an assessment of the practicality of a proposed project or system. It is conducted during the initiation phase to determine if the project is worth the investment.

Types of Feasibility:

1. Economic: Do we have the budget? what is the ROI (Return of Investment)?
2. Technical: Do we have the technology and expertise to build this?
3. Operational: Will the system be used effectively once it is built? Does it ~~solve~~ solve the business problem?
4. Legal: Does the project comply with data privacy laws and regulations?
5. Schedule: Can we finish the project in the required timeframe?

Purpose & Contents:

- The purpose is to provide management with the information needed to decide whether to "Go" or "No-Go". The report usually contains an executive summary, description of the current system, proposed solution and a cost/benefit analysis.

Q7 Write a short note on W.B.S

- The Work Breakdown Structure (WBS) is a deliverable-oriented hierarchical decomposition of the work to be executed by the project team.

Key Features:

- Feature: Hierarchical - It starts with the final project at the top and breaks it down into smaller, manageable components.
- Work Packages: The lowest level of the WBS is called a "work package," which can be easily scheduled and costed.
- The 100% rule: The WBS must include 100% of the work defined by the project scope - nothing more, nothing less.
- Visual Tool: It helps in identifying unnecessary tasks and prevents "scope-creep" (uncontrolled changes in project scope).